DPMN: Deep Prior Mamba Network for Hyperspectral Anomaly Detection

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***Abstract*—Recent advancements in hyperspectral anomaly detection (HAD) utilizing deep learning have garnered significant attention due to their superior performance. However, most existing methods based on convolutional neural networks (CNNs) and Transformer focus on extracting local and global features separately and assume that the background resides in a single subspace for reconstruction, thereby reducing the quality of the reconstructed background and decreasing the accuracy of HAD. Moreover, although incorporating the prior physical knowledge into the loss function can enhance the performance of the deep learning networks, it also increases the number of hyperparameters and complicates the tuning process. To address these issues, we propose a deep prior Mamba network (DPMN) for HAD, which primarily consists of two components: the abundance generation module (AGM) and the background reconstruction module (BRM). Specifically, AGM employs convolutional layers to extract local information and introduces Mamba to capture long-range dependencies, achieving feature extraction from local to global. Subsequently, BRM utilizes a learnable background dictionary to divide the background into multiple subspaces for reconstruction, realizing accurate background reconstruction while effectively suppressing the interference of anomalies on background reconstruction. Furthermore, to fully leverage the intrinsic properties of hyperspectral images (HSIs), we incorporate a regularization term into the loss function, merging the total variation (TV) with the low-rank representation (LRR), which not only exploits spatial smoothness and low ranknes but also reduces the number of hyperparameters. Experimental results on eight publicly available real datasets demonstrate that our method significantly outperforms other state-of-the-art (SOTA) methods. In addition, our code is available at: https://github.com/I3ab/DPMN**

***Index Terms*—Anomaly detection, hyperspectral images (HSIs), low-rank representation (LRR), Mamba, total variation (TV).**

I. Introduction

HYPERSPECTRAL images (HSIs), encompassing two spatial dimensions and one spectral dimension, are regarded as a 3-D cube. The spectral dimension is comprised of dozens to hundreds of narrowbands ranging from visible-to-near-infrared wavelengths, providing each pixel in HSIs with a nearly continuous spectral curve [1]. This allows for a more accurate reflection of the characteristics of surface materials. Owing to the inherent rich spatial and spectral information in HSIs, they are widely applied in areas such as classification, unmixing, and target detection.

Hyperspectral anomaly detection (HAD) aims to identify anomalous targets with significant differences from their surrounding background without prior knowledge of the target's spectral characteristics, belonging to unsupervised hyperspectral target detection [2]. In the real world, the anomaly detection is highly practical due to the difficulty in obtaining the target's spectral characteristics and the existence of spectral variability. HAD has been widely used to detect and monitor anomalies or changes in the environment, such as geological exploration, environmental monitoring, and agricultural management [3].

Generally, due to the low probability of anomalous targets appearing in a scene, anomalous targets exhibit sparse characteristics in space. In addition, the background of HSIs possesses spatial smoothness and low-rank properties. Fully utilizing these properties is highly beneficial for HAD.

Over the years, numerous methods for addressing HAD problems have emerged, primarily categorized into two types: physics-based methods and deep learning-based methods.

A. Physics-Based Methods

The statistical-based Reed–Xiaoli (RX) global method [4] is one of the most classic, often used as a benchmark algorithm. It is based on the assumption that the background obeys the multivariate Gaussian distribution and uses the Mahalanobis distance to determine the anomalies. The weighted RX [5] and kernel RX [6], [7], [8] have been proposed to enhance the performance of the RX method. In addition, many statistical-based methods have been proposed [9], [10], [11], [12], [13]. However, the assumption that the background follows a Gaussian distribution is unreliable in complex real-world scenarios, thereby limiting the detection accuracy of such methods.

To address this shortcoming, the sparse representation (SR)-based methods and the collaborative representation (CR)-based methods have been proposed, which do not rely on specific assumptions about the background distribution, but rather assume that background pixels can be linearly represented by a background dictionary, whereas anomalies cannot. In SR-based anomaly detection methods [14], [15], [16], [17], the main idea is to represent each background pixel as a sparse

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combination of a set of basic atoms from their spatially neighboring pixels. In contrast, the detectors based on CR [18], [19], [20] emphasize the collaborative ability of dictionary atoms to represent the background pixel. However, both SR-based detectors and CR-based detectors process pixels individually, failing to fully utilize the correlations between pixels. To make up for this shortcoming, the methods based on the low-rank decomposition model (LRDM) have gradually become an effective choice in recent years [21], [22].

Since the background occupies the majority of regions in HSIs and the number of background object types in the scene is far less than the number of spectral bands, the background of HSIs exhibits low-rank characteristics [23]. Thus, LRDM can be employed to effectively represent HSIs, capturing their essential characteristics, thereby enabling LRDM-based methods to more accurately distinguish between background pixels and anomalous pixels, leading to effective anomaly detection. LRDM-based methods primarily fall into two categories those based on: low-rank representation (LRR) and tensor LRR (TLRR).

The methods based on LRR typically incorporate the spatial dimension of the HSIs, transforming the image data into a 2-D matrix format for optimization. Among these methods, the spatial and spectral dimensions of HSIs are explicitly defined, with both dimensions considered separately during processing. It is worth noting that robust principal component analysis (RPCA) [24], [25], [26] is one of the classic LRDM, which also converts HSIs into 2-D matrices for optimal solving and effectively separates the observed data into low rank and sparse components. RPCA assumes that the observed data all lie in the same subspace for processing. However, HSIs containing multiple land cover types are more suitable to be divided into multiple subspaces for processing. The LRR-based methods recover the underlying multisubspace structure in data by obtaining an appropriate dictionary. To further enhance the performance of LRR-based methods, GTVLRR [27] introduces a total variation (TV) regularization term to exploit the spatial smoothness of the background and a graph regularization term to preserve the local geometric structure in the data. The LSCTV [28] exploits TV to obtain local spatial structures within the segmented super-pixel background blocks. The effective anomaly space (EAS) for HAD [29] incorporates sparsity cardinality and independent component analysis into LRR-based methods.

To preserve the inherent 3-D cube characteristics of HSIs during HAD, TLRR can be employed to capture the image structure. The method based on the tensor low-rank SR (TLRSR) [30] is proposed to prevent the intrinsic connections within HSIs from being disrupted. Moreover, the low average rank with the TV regularization method for AD (LARTVAD) [31] simultaneously imposes both TV and tensor low-rank constraints on 3-D HSIs. Furthermore, the tensor ring (TR) decomposition with factors TV regularization model for AD (TRDFTVAD) [32] extracts factor tensors on the background of 3-D HSIs and applies TV regularization to them, thereby achieving smooth and low-rank characteristics in both spatial and spectral dimensions. To further enhance the TV regularization, the multiscale entanglement renormalization ansatz (MERA) [33] tensor network proposes an HAD method based on MERA decomposition. Although the TLRR-based methods can capture complex 3-D data structures, they typically come with a high computational complexity. Moreover, tensor-based approaches usually treat all dimensions of observed data indiscriminately, processing each dimension identically. However, the two spatial dimensions and one spectral dimension of HSIs possess distinct physical meanings. Fully considering the differing physical significance of the spatial and spectral dimensions in HSIs will be beneficial in constructing HAD algorithms with the superior performance.

In general, the above methods integrate the physical knowledge, enhancing the model's physical interpretability and generalization capabilities, achieving better detection performance even with the limited data. However, selecting an appropriate model to accurately describe various complex HSIs scenarios in practice remains challenging, which significantly limits the detection accuracy. In addition, since most physics-based methods require iterative optimization for detection, they incur prolonged runtime, making real-time detection difficult to achieve.

B. Deep Learning-Based Methods

In recent years, with the rapid development of deep learning, numerous HAD methods based on deep learning have emerged to improve the effectiveness and efficiency of detection. The deep learning-based methods are primarily categorized into two types: unsupervised methods and self-supervised methods.

HAD is an unsupervised task, making unsupervised methods particularly suitable for HAD. In recent years, a variety of unsupervised methods based on autoencoders (AEs) and Transformers have been proposed successively. The AE-based method for HAD (Auto-AD) [34] obtains the latent features of the background through the encoder, then reconstructs the background by using the decoder, and ultimately identifies anomalous targets through the reconstruction loss. The robust graph AE (RGAE) [35] and the guided AE-based detector (GAED) [36] integrate graph constraints and guidance modules to enhance the performance of AEs, respectively. The fully convolutional AE for HAD [37] incorporates a spatial–spectral joint attention mechanism to improve the utilization of spatial information. Due to the powerful reconstruction capabilities of AEs, backgrounds can be effectively reconstructed. However, this may also result in anomalies being mistakenly reconstructed as part of the background, leading to the failure to detect certain anomalies. The deep feature aggregation network (DFAN) for HAD [38] employs an adaptive clustering network to cluster background and anomalies separately, and designs a multiaggregation separation loss to enhance the reconstruction of background and suppress the interference of anomalies on background reconstruction. However, the presence of an excessive number of parameters in the AEs may result in overfitting. Recently, an AE architecture based on skip connection was proposed [39] for hyperspectral unmixing, which reduces the number of layers of the traditional AE, and thus, the number of parameters of the network. Notably, the aforementioned methods are

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all based on a convolutional neural network (CNN), which have limited receptive fields and struggle to capture long-range dependencies between pixels.

The Transformer-based methods can be utilized to solve the above problems. The spatial–spectral Transformer network (SSTR) [40] leverages two cascaded Transformer to mine the rich spatial and spectral information in HSIs, respectively. The Transformer with spatial–spectral dual-window mask for HAD (S2DWMTrans) [41] utilizes both global and local information for background reconstruction and designs an adaptive loss function to suppress the interference of anomalies in background reconstruction. The gated Transformer for HAD (GT-HAD) [42] designs two modules, respectively, to enhance the background reconstruction, and it also designs a gating mechanism for control selection. However, the attention mechanism in Transformer results in computational complexity that is proportional to the square of the sequence length.

Recently, some self-supervised methods for HAD have been proposed. The blind-spot self-supervised learning network (BS^3LNet) [43] first employs the concept of self-supervised learning to solve the HAD task, and it uses a blind-spot network to reconstruct the background. To improve the ability to distinguish anomalies from the surrounding background, the pixel-shuffle downsampling (PD) blind-spot reconstruction network (PDBSNet) [44] introduces the PD strategy for HAD. Furthermore, the multiscale mask convolution-based blind-spot network (MMBSN) [45] employs the multiscale convolution to improve the capability of the blind-spot network in handling anomalies of different sizes.

In general, deep learning-based methods can automatically learn task-appropriate feature representations from data, accompanied by shorter inference computation times, making them more suitable for real-time detection. However, these methods often lack physical interpretability. While they generally offer higher accuracy compared with physics-based approaches, their generalization capabilities are relatively weaker. Furthermore, existing HAD methods fail to effectively integrate both local and global information and reconstruct the background across multiple subspaces, which may result in a reduction in detection accuracy. Although recent studies [46], [47], [48] have integrated prior physical knowledge into the loss functions of deep learning networks to enhance their physical interpretability, this integration significantly increases the number of hyperparameters and the complexity of tuning, thereby reducing the practical applicability of these methods.

C. Motivation and Contribution

To overcome the drawbacks of both physics-based and deep learning approaches, we propose the deep prior Mamba network (DPMN) for HAD, which falls under the category of an unsupervised network. The network primarily consists of two components: an abundance generation module (AGM) and a background reconstruction module (BRM). Specifically, we employ a skip-connection-based encoder–decoder structure to design AGM, which reduces the network size and consequently the number of network parameters. Given that CNN struggles to capture global information between pixels, the Mamba [49] block is introduced, which integrates state-space models (SSMs) with deep learning to efficiently capture long-range dependencies within sequences. To comprehensively capture long-range dependencies between pixels, we integrate a bidirectional Mamba block into AGM, enabling more effective extraction of background features and the restoration of a more precise background. In BRM, we incorporate a background dictionary, enabling the background to be divided into different subspaces, thereby facilitating a more accurate reconstruction of the background. Furthermore, since anomalies cannot be represented linearly by background dictionaries, it is difficult to reconstruct anomalies in BGM, thereby reducing the likelihood of anomalies appearing in the reconstructed background, which enhances the detection accuracy. In addition, to incorporate the general prior physical knowledge of HSIs, we add a regularization term that merges TV into LRR in the loss function, which fully utilizes the most important properties of HSIs (spatial smoothness and low rankness) to guide the network in reconstructing more accurate background, while introducing only one hyperparameter, which reduces the interaction between the regularization terms and the difficulty of hyperparameter tuning compared with using these two properties alone.

Comprehensive experiments are conducted on eight real-world datasets with varying anomaly sizes to validate the effectiveness and generalization capability of the proposed method in enhancing the detection performance compared with the current state-of-the-art (SOTA) approaches. Moreover, a series of ablation experiments is designed to verify the effectiveness of the network components, the convenience of parameter tuning, and the noise robustness of the network.

The main contributions of this article can be concisely summarized as follows.

1) We construct an AGM to generate background abundance, which not only utilizes convolutional layers to extract local features but also adopts a Mamba block that scans bidirectionally to capture global features, facilitating the subsequent recovery of a more accurate background.
2) To divide the background into different subspaces for reconstruction while reducing the interference of anomalies on background reconstruction, we integrate the background dictionary and generated abundances into the BRM, which combines them through a dictionary-learnable block to recover the background. Furthermore, a convolutional layer is incorporated to ensure precise restoration of image edge structures and smooth background, leading to the final accurately reconstructed background.
3) We incorporate a regularization term that merges TV into LRR in the loss function, which reduces the mutual constraints between regularization terms and decreases the number of hyperparameters and their tuning difficulty, thereby effectively guiding the network toward accurate background reconstruction. Moreover, benefiting from the incorporation of prior physical knowledge, the robustness and generalization capabilities of the network can be significantly enhanced.

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The remainder of this article is organized as follows. The basic concept of SSMs and density peak-based clustering (DPC) algorithm is reviewed in Section II. The proposed method is presented in Section III, and its effectiveness is demonstrated by the experimental results on the eight datasets in Section IV. The conclusion can be found in Section V.

II. Related Works

A. State-Space Models

To address the shortcomings of CNNs in capturing global information, Transformers have been applied [50], [51]. Their self-attention mechanism excels in handling long-range dependencies, but the computational complexity of the self-attention mechanism is proportional to the square of the sequence length, making it computationally expensive for long sequences. The structured SSM [52], [53] is an architecture that combines SSM with deep learning, designed to effectively handle long-sequence data tasks. By leveraging the advantages of state-space dynamic modeling, SSM can capture long-term dependencies in sequences while maintaining the efficient computation and storage. However, traditional SSM employs fixed structured modeling and does not support attention mechanisms.

To further enhance the performance of SSM, Mamba [49] designs a state space selective mechanism, making the model more flexible. Specifically, Mamba enhances its capability to model both local patterns (such as short-term changes) and global dynamics by adding convolutional layers. Moreover, Mamba enables the learning of complex sequence patterns by incorporating nonlinear activation layers, rather than being restricted to linear transformations.

The advantages of Mamba over Transformer include higher computational efficiency [54] (complexity close to $O(N)$), lower memory requirements, stronger capabilities in modeling long-term and short-term dependencies, and flexibility in dynamic parameterization, enabling it to accommodate nonstationary sequences and dynamically changing inputs. Furthermore, Mamba is more robust to noisy data and small-scale datasets, making it suitable for scenarios with limited resources or high real-time requirements. However, in Mamba, achieving scanning across a 2-D space constitutes a pivotal step, necessitating the transformation of pixel data from a 2-D grid into a 1-D sequence of pixels. Nevertheless, relying solely on a single scanning direction might constrain Mamba's ability to effectively the capture global information. Currently, Mamba has been applied in various fields, including change detection [55], image segmentation [56], and classification [57].

B. Density Peak-Based Clustering

The DPC algorithm [58], [59] performs clustering based on the similarity between pixel points. The category center has two properties: 1) the local density of sample points around the category center is lower than it and 2) the distance between the category center and all points with higher density than it is larger. Specifically, for each pixel i, the local density ρ_i is calculated according to (1). After confirmation, the minimum distance δ_i between the pixel and other sample points with higher density than it is calculated according to (2)

$$\rho_i = \sum_{j=1, j\neq i}^{n} \exp\left(-\left(\frac{d_{i,j}}{d_c}\right)^2\right) \tag{1}$$

$$\delta_i = \min_{j:\rho_j>\rho_i}\left(d_{i,j}\right) \tag{2}$$

where d_c represents the cutoff distance, which is usually calculated by

$$d_c = \mathbf{d}_s(p), \quad p = \lfloor \alpha \times n \times (n-1) \rfloor \tag{3}$$

where n is the number of samples, $\mathbf{d}_s$ represents the ascendingly sorted results of all $d_{i,j}(i \neq j)$. Also, α serves as the scaling factor, determining the percentage of samples considered as neighbors, which in turn influences the value of d_c and consequently affects the number of clusters. The clustering results are robust to the value of d_c, and α is typically set at 2%, as recommended in [58].

To automatically determine the number of categories and their corresponding category centers, before calculating the score, ρ_i and δ_i should be normalized, and the score of each pixel i is calculated by (4), and the number of categories K is confirmed according to (5), and the scores are then sorted in descending order, with the top K scores selected as the category centers. Once the center of the category is determined, the remaining cells are assigned to the class in which the nearest neighbor cell is located with a higher density than their own

$$\gamma_i = \rho_i \times \delta_i^2 \tag{4}$$

$$\left|\lg\left(\gamma_{K+1}\right) - \lg\left(\gamma_{K+2}\right)\right| < \beta, \quad \left|\lg\left(\gamma_{K+2}\right) - \lg\left(\gamma_{K+3}\right)\right| < \beta \tag{5}$$

where β is a preset threshold, empirically set to 0.1, according to [59].

The DPC algorithm effectively achieves favorable clustering results, characterized by its ability to automatically determine the number of clusters without manual parameter setting. In addition, the algorithm can identify points of lower density, which typically represent outliers.

III. Proposed Method

Usually, the HAD methods based on deep learning primarily detect anomalous targets by subtracting the reconstructed background from the original imagery. Therefore, the quality of background reconstruction plays a crucial role in enhancing the HAD accuracy. However, the previous methods have failed to effectively integrate local and global information and reconstruct the background within multiple subspaces, which may result in a decrease in detection accuracy. Therefore, we propose DPMN for HAD, which effectively integrates local and global information and utilizes a background dictionary to divide the background into distinct subspaces, where the dictionary is generated by a modified DPC algorithm. Specifically, the proposed method consists of the following modules: AGM for generating abundances and the BGM for reconstructing the background by means of a dictionary-learnable block, as shown in Fig. 1.

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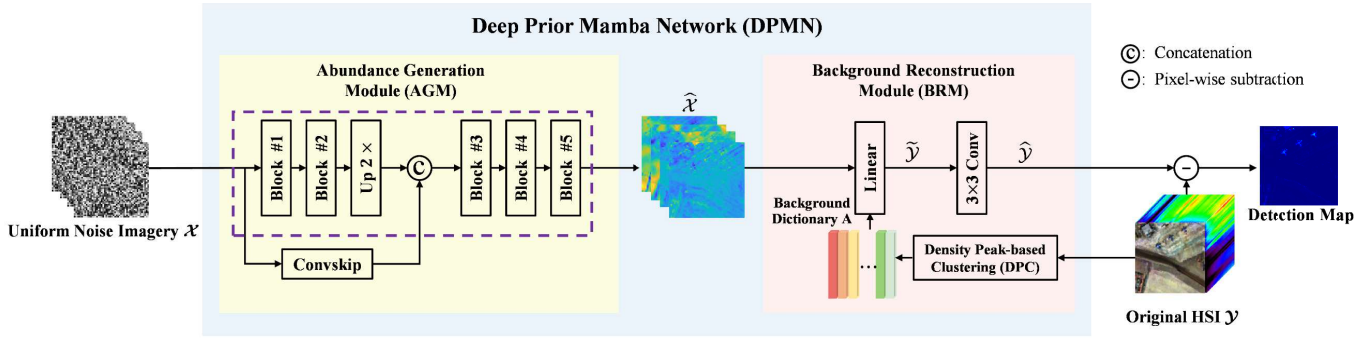


Fig. 1. Overall framework of the proposed method.

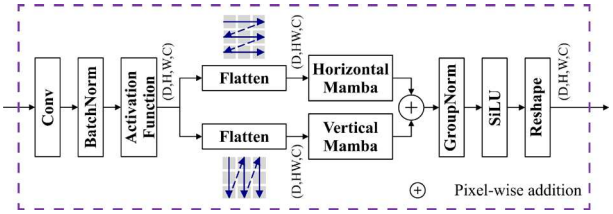


Fig. 2. Structure of Blocks #1–#5 in AGM.

The input of the network is a uniform noise imagery with the same size as the abundance of the background. First, the AGM generates the abundances. Next, the generated abundances are combined with the background dictionary within the BGM to reconstruct the background component of HSIs. Finally, the anomalous detection map is obtained by subtracting the reconstructed background from the original imagery.

Furthermore, a loss function containing reconstruction loss and a regularization term that merges TV into LRR is designed, aiming to fully leverage the spatial smoothness and low-rank properties of HSIs.

A. Abundance Generation Module

AGM consists of five convolutional layers, one convolutional skip layer, and three bidirectional Mamba, with the overall framework illustrated in Fig. 1. The input data are a uniform noise imagery $\mathcal{X} \in \mathbb{R}^{H\times W\times r}$ of the same size as the background component, where H and W represent the height and width of the original imagery, respectively, and r represents the atomic number of the dictionary. Similar to methods based on CNN, we employ convolutional layers to extract local information between pixels. Obviously, the convolutional skip connection layer not only prevents gradient vanishing and explosion but also enhances the efficiency of information transmission by extracting features. The structures of Blocks #1–#5 in AGM are identical, as shown in Fig. 2, and the detailed configurations of the convolutions and activation functions within each block are specified in Table I. The Softmax activation function is adopted in the AGM due to the nonnegativity and sum-to-one properties of abundances.

To better learn the characteristics of the background, we incorporate the Mamba block into AGM. Since the Mamba block operates by transforming images into sequences, we adopt the bidirectional Mamba block to comprehensively capture long-range relationships between pixels, as illustrated in Fig. 2. This allows the network to learn the distribution of background features more effectively, thereby enabling the precise separation of background from anomalies. Moreover, since Mamba incorporates a state space selective mechanism and employs parallel algorithms that take hardware characteristics into account, the efficiency of network operations is enhanced. The specific mathematical expression is as follows:

$$\widetilde{\mathcal{X}}_H = \text{Reshape}\left(\text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\widetilde{X}\right)\right)\right)\right) \tag{6}$$

$$\widetilde{\mathcal{X}}_V = \text{Reshape}\left(\text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\widetilde{X}\right)\right)\right)\right) \tag{7}$$

$$\widetilde{\mathcal{X}} = \widetilde{\mathcal{X}}_H + \widetilde{\mathcal{X}}_V \tag{8}$$

where $\widetilde{X} \in \mathbb{R}^{HW\times C}$ is the output of the flatten layer and $\widetilde{\mathcal{X}}_H \in \mathbb{R}^{H\times W\times C}$ and $\widetilde{\mathcal{X}}_V \in \mathbb{R}^{H\times W\times C}$ represents the output of the horizontal Mamba and vertical Mamba, respectively.

TABLE I
DETAILED CONFIGURATION OF BLOCKS #1–#5 AND CONVSKIP IN AGM

Block No.	Convolution	Activation Function
Block #1	Conv2d(r-128, 3×3, stride 2)	LeakyReLU
Block #2	Conv2d(128-128, 3×3, stride 1)	LeakyReLU
Block #3	Conv2d(132-128, 3×3, stride 1)	LeakyReLU
Block #4	Conv2d(128-128, 1×1, stride 1)	LeakyReLU
Block #5	Conv2d(128-r, 1×1, stride 1)	Softmax
Convskip	Conv2d(r-4, 1×1, stride 1)	LeakyReLU

B. Background Reconstruction Module

The BRM combines the abundances generated by AGM with a prior dictionary through a dictionary-learnable block, which helps to recover a more accurate background.

The prior dictionary is generated through the DPC algorithm. Specifically, the DPC algorithm is first employed to extract the cluster centers. Then, to enhance the stability and robustness of the background dictionary, apart from the central pixel, the top $S-1$ pixels are also chosen based on the descending order of the scores in (4), which forms the background dictionary $\boldsymbol{A}$ together with the category center.

During the background reconstruction, the generated abundances are initially fed into a linear layer. The obtained background dictionary $\boldsymbol{A}$ is used for parameter initialization of the linear layer, meaning that the linear layer serves as a

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dictionary-learnable block, allowing the background dictionary $\boldsymbol{A}$ to be adjusted and optimized through network training and then combined with the abundances to obtain the reconstructed background. However, the linear layer can only reconstruct most of the background information by combining input abundances $\widehat{\mathcal{X}}$ with a background dictionary $\boldsymbol{A}$, with some edge information easily being overlooked. Therefore, we add a convolutional layer (spatial smoothing filter) to enable the network to better capture the edge information and smooth the background, which aligns with the spatial smoothness of the background, thereby ultimately improving the accuracy of background reconstruction. The specific mathematical expression is as follows:

$$\widetilde{\mathcal{Y}} = \text{Linear}\left(\widehat{\mathcal{X}}, \boldsymbol{A}\right) \tag{9}$$

$$\widehat{\mathcal{Y}} = \text{Conv}_{3\times 3}\left(\widetilde{\mathcal{Y}}\right) \tag{10}$$

where $\widehat{\mathcal{X}} \in \mathbb{R}^{H\times W\times r}$ is the abundances generated by AGM, $\boldsymbol{A}$ is the preextracted dictionary, r is the atomic number of the dictionary, and $\widehat{\mathcal{Y}} \in \mathbb{R}^{H\times W\times B}$ is the final reconstructed background.

C. Loss Function

The loss function used during the training phase consists of two parts: one is a weighted reconstruction loss ($\mathcal{L}_{\text{wr}}$) designed to promote background reconstruction while suppressing the interference of anomalies on background reconstruction, and the other is a regularization term ($\mathcal{L}_{\text{tvlrr}}$) that merges TV into LRR to prevent overfitting and enhance the generalization capabilities of the network.

$\mathcal{L}_{\text{wr}}$ achieves the goal of encouraging background reconstruction while reducing the interference of anomalies on background reconstruction by imposing a large weight on background pixels and a small weight on potentially anomalous ones. The specific calculation is as follows:

$$\mathcal{L}_{\text{wr}} = \frac{1}{N}\sum_{i}^{H}\sum_{j}^{W}\left(\left(\mathcal{Y}_{i,j,:} - \widehat{\mathcal{Y}}_{i,j,:}\right) w_{i,j}\right)^2 \tag{11}$$

where the calculation of weights is obtained through reconstruction error $S = [s_{1,1}, \ldots, s_{1,W}, \ldots, s_{H,1}, \ldots, s_{H,W}]$, similar to [34], with the specific calculation as follows:

$$s_{i,j} = \left\|\mathcal{Y}_{i,j,:} - \widehat{\mathcal{Y}}_{i,j,:}\right\|_2^2 \tag{12}$$

$$\widehat{w}_{i,j} = \max(S) - s_{i,j} \tag{13}$$

$$w_{i,j} = \frac{\widehat{w}_{i,j} - \min\left(\widehat{w}_{i,j}\right)}{\max\left(\widehat{w}_{i,j}\right) - \min\left(\widehat{w}_{i,j}\right)}. \tag{14}$$

$\mathcal{L}_{\text{tvlrr}}$ leverages two crucial properties of the background, namely, spatial smoothness and low rankness. The TV regularization term encourages the network to produce smooth HSIs, thereby reducing the image noise and suppressing the reconstruction of anomalies. LRR promotes HSIs with lower ranks, improving the robustness. Moreover, the integration of TV and LRR as a priori physical knowledge can enhance the generalization capabilities of the network. To simultaneously utilize TV and LRR while reducing the number of hyperparameters, we construct a regularization term merging TV into LRR, which is specifically written as follows:

$$\mathcal{L}_{\text{tvlrr}} = \left\|\text{TV}\left(\widehat{\mathcal{X}}\right)\right\|_* = \left\|\Delta_h\widehat{\mathcal{X}}\right\|_* + \left\|\Delta_v\widehat{\mathcal{X}}\right\|_* \tag{15}$$

where $\|\cdot\|_*$ denotes the nuclear norm. Note that when calculating the nuclear norm of a tensor, we first unfold the tensor into a 2-D matrix, i.e., unfolding a tensor of dimension $H \times W \times r$ into a matrix of dimension $HW \times r$. TV($\widehat{\mathcal{X}}$) is the differential tensor along horizontal and vertical directions, where Δ_h and Δ_v denote the respective operations, with the calculation formula illustrated as follows:

$$\begin{cases}\Delta_h\widehat{\mathcal{X}}(i,j,k) = \widehat{\mathcal{X}}(i+1,j,k) - \widehat{\mathcal{X}}(i,j,k)\\ \Delta_v\widehat{\mathcal{X}}(i,j,k) = \widehat{\mathcal{X}}(i,j+1,k) - \widehat{\mathcal{X}}(i,j,k).\end{cases} \tag{16}$$

Finally, the total loss can be written as follows:

$$\mathcal{L} = \mathcal{L}_{\text{wr}} + \lambda\mathcal{L}_{\text{tvlrr}} \tag{17}$$

where λ is the only hyperparameter in the loss function, which is used to adjust the impact of regularization on the loss function.

In the testing phase, the test dataset $\mathcal{Y}$ is input into the trained network. The network outputs the reconstructed background while suppressing the interference of anomalies on background reconstruction. Consequently, the pixels with large reconstruction errors are regarded as anomalies, and the anomalous degree of each pixel is calculated by the following equation:

$$D_{i,j} = \left\|\mathcal{Y}_{i,j,:} - \widehat{\mathcal{Y}}_{i,j,:}\right\|_2^2. \tag{18}$$

IV. Experimental Results and Analysis

In this section, we first compare and evaluate the proposed method against the current SOTA methods on eight commonly used datasets, which cover different sizes of anomalous targets. Subsequently, we conduct ablation experiments and analysis, including component ablation, parameter analysis, and noise robustness experiments, to assess the contribution of each component to the overall performance, explore the impact of different parameter settings on the results, and investigate the robustness and performance stability of the algorithm under various noise conditions. Finally, we analyze the complexity and convergence of the proposed network.

A computer with an Intel Core i9, 3.50-GHz CPU, 32.0-GB RAM, and a single RTX 3090 GPU with 24 GB of memory serves as a platform to implement all experiments. The physics-based methods are implemented on MATLAB 2023b, and the deep learning-based methods and the proposed DPMN are implemented on the Pytorch toolbox.

A. Hyperspectral Datasets

We selected eight commonly used real datasets to evaluate the detection performance of the proposed method as well as the comparative methods. Given that anomaly detection methods exhibit varying performance in detecting anomalous targets of different sizes [60], particularly in practical applications where precisely identifying small and subpixel anomalies poses greater challenges, the chosen datasets encompass

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Fig. 3. (Top) Pseudo-color images and (bottom) ground truth of the eight real datasets. (a) HYDICE. (b) Hyperion. (c) ABU-beach-1. (d) ABU-beach-3. (e) Pavia. (f) Salinas. (g) AVIRIS-1. (h) AVIRIS-2.

anomalies of various sizes. This selection aims to comprehensively assess the detection capabilities of the proposed method, thereby providing a more authentic representation of its effectiveness and applicability in real-world scenarios. Fig. 3 displays the pseudo-color images of various datasets along with their corresponding ground truth labels, with the size of anomalous targets increasing from left to right.

HYDICE [61]: This real dataset is a 100× 100 subimagery used for experiments, which is cropped from the original imagery with a size of 400 ×400 and 174 bands. The original imagery was obtained by the HYDICE sensor. The vehicles in the dataset are considered anomalous targets and have 21 pixels.

Hyperion [62]: This real dataset was acquired by the Hyperion hyperspectral imaging spectrometer aboard the Earth Observing-1 (EO-1) satellite of the National Aeronautics and Space Administration (NASA). Its size is 150 × 150, and there are 155 bands left after excluding the bands of the low signal-to-noise ratio (SNR). The storage silos in this dataset are considered as anomalous targets and have 17 pixels.

ABU-Beach-1 and ABU-Beach-3 [63]: These real datasets were collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor at the Cat Island and the Bay Champagne, respectively. They all possess 188 bands. The ABU-beach-1 dataset is 150 × 150 in size and has 19 anomalous pixels. The ABU-beach-3 dataset has a size of 100 × 100 and contains 11 anomalous pixels. The anomalous targets in the ABU-beach-1 and ABU-beach-3 datasets are aircraft and ships, respectively.

Pavia [64]: This dataset has 102 bands with a size of 100×100, which is taken from the original imagery with a size of 1096 ×1096 obtained by the ROSIS sensor in Pavia, Northern Italy. The anomalous targets to be identified are vehicles in the dataset, which contains 71 anomalous pixels.

Salinas [65]: This real dataset was collected by the AVIRIS sensor. The size of the original imagery is 512 × 217, with 224 bands. The experimental dataset is a subimagery of size 200 × 150, with 204 bands after removing severely noise-affected bands. The anomalous targets within the dataset are man-made constructs, occupying 42 pixels.

AVIRIS-1 and AVIRIS-2 [66]: These real datasets were obtained by the AVIRIS sensor from the San Diego airport. Both are of size 100 × 100, containing 186 and 189 bands, respectively. The planes in the scenes are anomalous targets, occupying 72 pixels and 77 pixels in the two datasets, respectively.

B. Evaluation Metrics

To comprehensively and effectively compare the detection accuracy of the different approaches, the 3-D receiver operating characteristic (3-D ROC) curve proposed in [67] for HAD is employed as the evaluation metric. In addition, the 2-D ROC(P_D, P_F) and 2-D ROC(P_F, τ) curves plotted from the 3-D ROC curves, where P_D and P_F are the probabilities of detection and false alarm, respectively. In addition, $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, $\text{AUC}_{(F,\tau)}$, AUC_{ODP}, and AUC_{SNRP} are used as evaluation metrics. Here, $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$ represent the area under the ROC(P_D, P_F), ROC(P_D, τ), and ROC(P_F, τ) curves, respectively.

The AUC_{ODP} and AUC_{SNRP} are calculated by the following equations, respectively:

$$\text{AUC}_{\text{ODP}} = \left(1 - \text{AUC}_{(F,\tau)}\right) + \text{AUC}_{(D,\tau)} + \text{AUC}_{(D,F)} \quad (19)$$

$$\text{AUC}_{\text{SNRP}} = \frac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}}. \quad (20)$$

To compare the computational cost of different methods, the running time of each method is recorded. The physics-based methods require dozens to hundreds of iterations for each result acquisition. However, for the deep learning-based methods, since the network can obtain results directly through testing after training, both the training time and testing time are provided for comparison.

C. Comparison With Other Methods

In this section, we compare the proposed method with recent advanced physics-based approaches, including PCA-TLRSR [30], LARTVAD [31], and TRDFTVAD [32], which are capable of preserving the intrinsic 3-D structure of HSIs and leveraging the characteristics of HSIs. In addition, we contrast our method with four SOTA deep learning-based approaches: Auto-AD [34], RGAE [35], GAED [36], GT-HAD [42], and DFAN [38]. In the experiments across all datasets, the unique hyperparameter λ of the proposed method is set to 0.001. Also, the code for PCA-TLRSR [30], LARTVAD [31], TRDFTVAD [32], Auto-AD [34], RGAE [35], GAED [36], GT-HAD [42], and DFAN [38] is available from their respective authors, and all parameter settings follow their corresponding References to ensure a fair comparison.

Fig. 4 shows the 2-D detection results of various methods. Obviously, although the anomalous targets detected by the proposed method do not look particularly prominent in the detection map, a significant advantage in background

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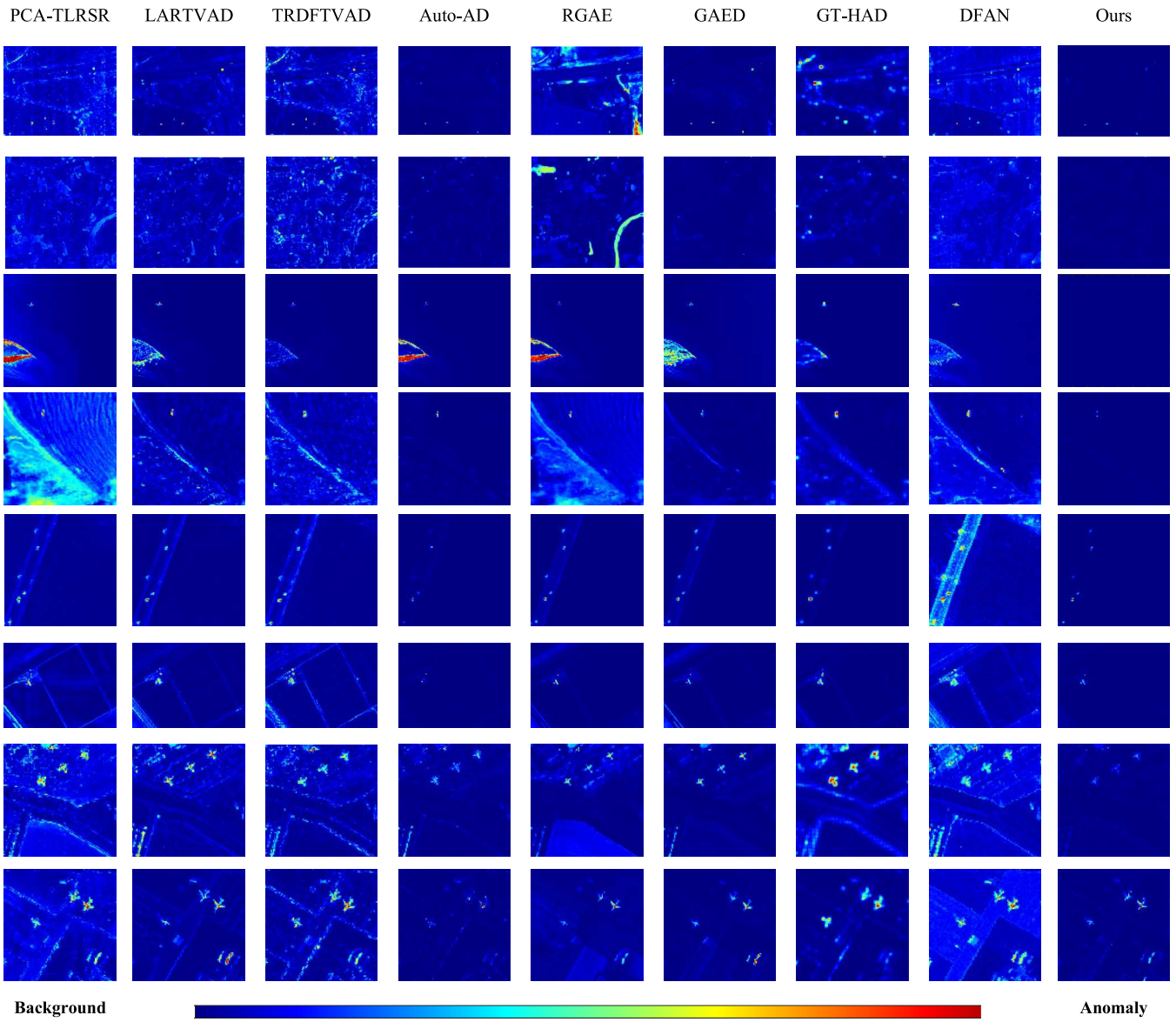


Fig. 4. Detection results of different methods on the eight real datasets. (Top to bottom) HYDICE, Hyperion, ABU-beach-1, ABU-beach-3, Pavia, Salinas, AVIRIS-1, and AVIRIS-2.

suppression is demonstrated, and the anomalous targets can also be identified. Overall, the proposed method demonstrates significant enhancement in background suppression compared with the comparative methods, thereby reducing the false alarm rate.

Figs. 5–7 display the 3-D ROC curves, ROC(P_D, P_F) and ROC(P_F, τ) curves, respectively. From Fig. 5, it can be observed that the curve of DPMN is closest to the top-right corner, indicating the best detection accuracy of DPMN. From Fig. 6, it can be seen that on the HYDICE, Hyperion, ABU-beach-3, Salinas, and AVIRIS-2 datasets, the detection rate of DPMN exceeds all comparison methods at nearly all false alarm rate values. On the ABU-beach-3, Pavia, and AVIRIS-1 datasets, when the false alarm rate is less than 10^{-2}, the detection rate of DPMN is lower than those of certain methods in LARTVAD, RGAE, GAED, GT-HAD, and DFAN. However, from the overall ROC(P_D, P_F) curve, our method still achieves the highest detection accuracy. Moreover, as shown in Fig. 6, the proposed method exhibits a rapid increase in P_D across all datasets within a lower range of P_F, reaching close to 1 earliest, indicating that other methods require a higher rate of false alarms to achieve the detection rate of the proposed method. In addition, as can be seen from Fig. 7, the curve of DPMN is closest to the bottom-left corner, indicating that DPMN has the best background suppression capability. In summary, the proposed method DPMN achieves excellent background suppression capabilities and detection accuracy.

The background-anomaly separation box maps are shown in Fig. 8, where the red box and blue box represent 10%–90% of the corresponding values of the anomalous targets and background component, respectively, and the dashed lines above and below the boxes indicate the remaining values. As can be observed from Fig. 8, the proposed method exhibits satisfactory capability in separating background and anomalies, as well as achieving the best background reconstruction performance.

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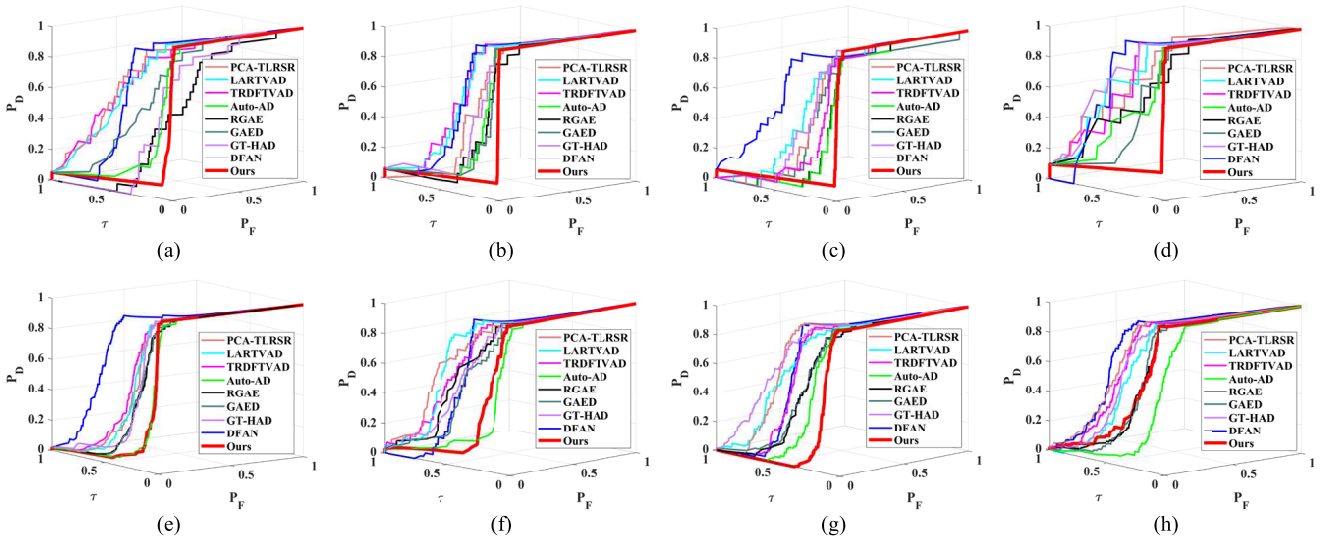


Fig. 5. 3-D ROC of different methods on the eight datasets. (a) HYDICE. (b) Hyperion. (c) ABU-beach-1. (d) ABU-beach-3. (e) Pavia. (f) Salinas. (g) AVIRIS-1. (h) AVIRIS-2.

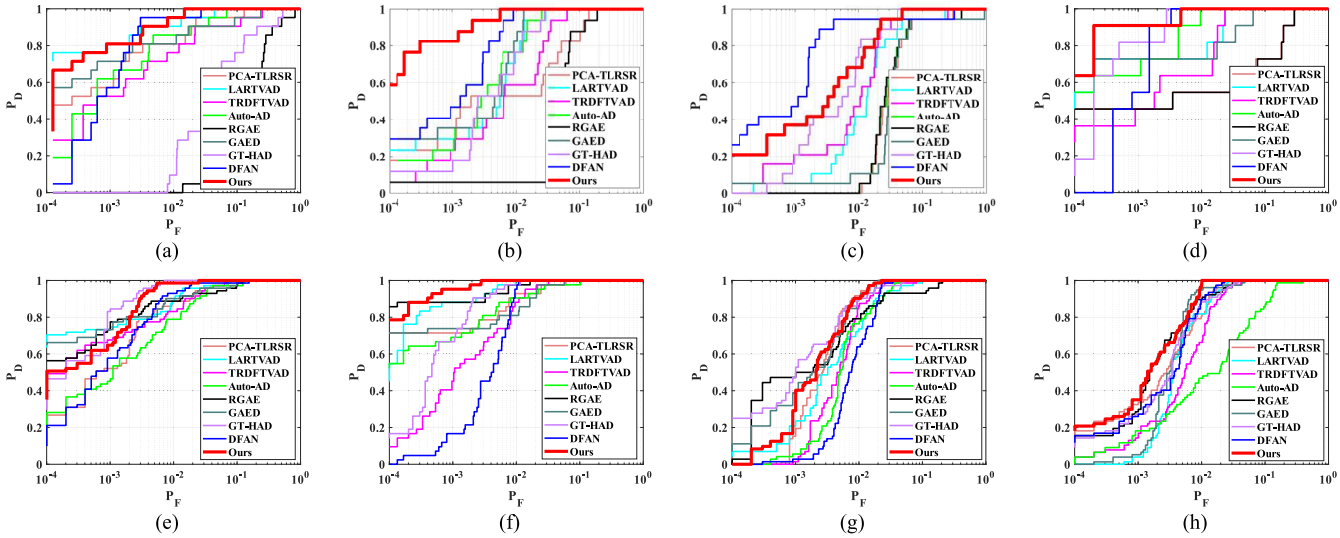


Fig. 6. ROC(P_D, P_F) curves of different methods on the eight datasets. (a) HYDICE. (b) Hyperion. (c) ABU-beach-1. (d) ABU-beach-3. (e) Pavia. (f) Salinas. (g) AVIRIS-1. (h) AVIRIS-2.

Table II provides five specific AUC values for different methods on different datasets, which can quantitatively demonstrate the performance of each method. As can be seen from Table II, the proposed method exhibits the highest $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{(F,\tau)}$, and $\mathrm{AUC}_{\mathrm{SNRP}}$ values, indicating the most comprehensive performance and superior background suppression capabilities. In addition, it can be seen that $\mathrm{AUC}_{(D,\tau)}$ and $\mathrm{AUC}_{\mathrm{OADP}}$ of the proposed method are lower than those of PCA-TLRSR, LARTVAD, and TRDFTVAD. This should be due to the inferior background reconstruction capabilities of PCA-TLRSR, LARTVAD, and TRDFTVAD, which result in a higher false positive rate, making them prone to mistakenly classify the background as the target. In terms of runtime, since the deep learning networks can directly output the inferred results for HAD after being trained, and the physics-based algorithms must be iterated to obtain the detection results, the runtime of methods based on deep learning is divided into training time and test time, while physics-based methods only have test time. As shown in Table II, the proposed method demonstrates significantly shorter test times compared with GT-HAD, thereby confirming that Mamba significantly outperforms Transformer in terms of computational efficiency. In addition, Table II indicates that the proposed DPMN has a relatively short test time, which should be suitable for achieving real-time detection.

Based on the experimental results of eight real datasets, the proposed method demonstrates the highest detection accuracy, with a relatively short test time.

D. Ablation Analysis

1) Component Ablation: Since the image data needs to be converted into sequences to be input into the Mamba block

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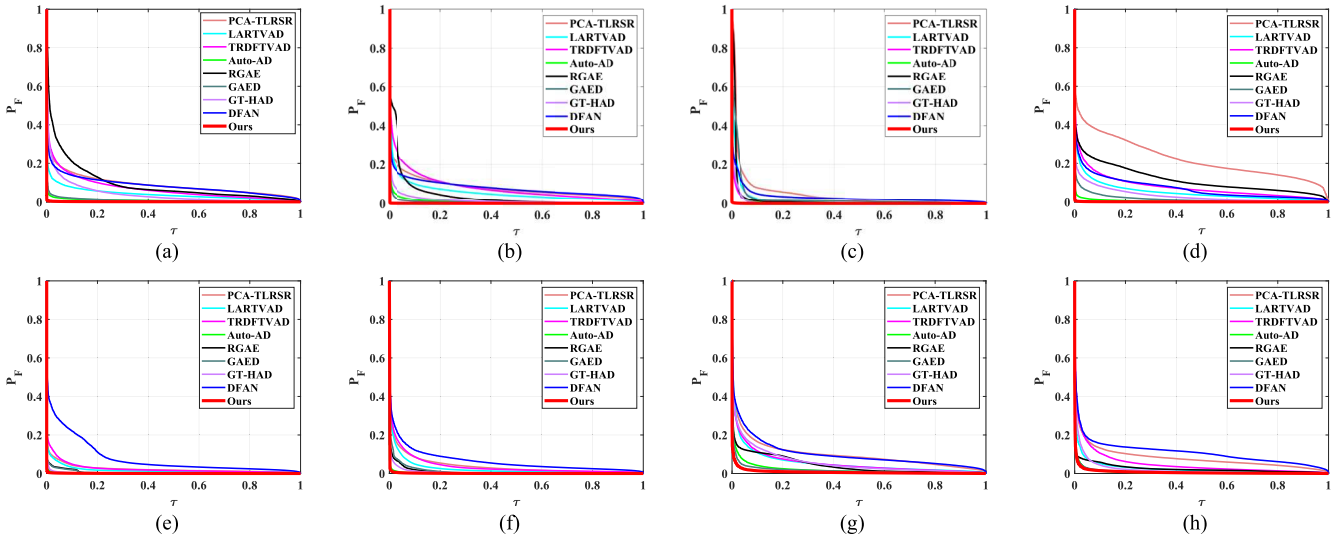


Fig. 7. ROC(P_F, τ) curves of different methods on the eight datasets. (a) HYDICE. (b) Hyperion. (c) ABU-beach-1. (d) ABU-beach-3. (e) Pavia. (f) Salinas. (g) AVIRIS-1. (h) AVIRIS-2.

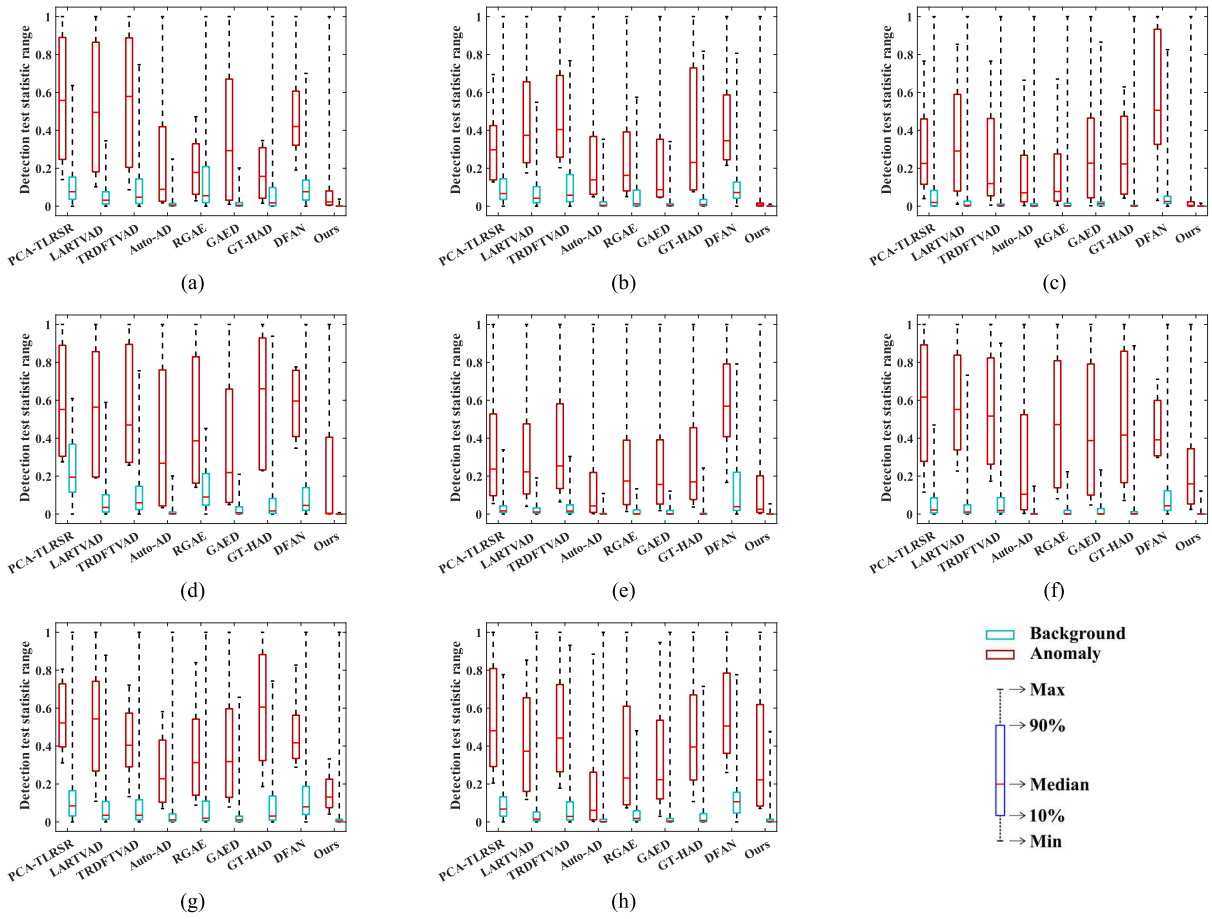


Fig. 8. Background-anomaly separation maps of different methods on the eight real datasets. (a) HYDICE. (b) Hyperion. (c) ABU-beach-1. (d) ABU-beach-3. (e) Pavia. (f) Salinas. (g) AVIRIS-1. (h) AVIRIS-2.

to obtain global information, the proposed method employs vertical and horizontal Mamba blocks to comprehensively establish relationships between different pixels, as shown in Fig. 9(a) and (b), respectively. In addition, sequences can also be constructed from a diagonal direction, as shown in Fig. 9(c). Table III shows the detection performances both with and without the diagonal direction. As shown in Table III, the introduction of diagonal direction not only does not significantly improve the performance but also increases the running time.

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TABLE II

FIVE AUC VALUES AND RUNNING TIME FOR DIFFERENT METHODS ON THE EIGHT DATASETS. BOLD AND UNDERLINE REPRESENT THE OPTIMAL AND SUBOPTIMAL, RESPECTIVELY

Datasets	AUC Values		PCA-TLRSR [30] (2023)	LARTVAD [31] (2023)	TRDFTVAD [32] (2023)	Auto-AD [34] (2022)	RGAE [35] (2022)	GAED [36] (2022)	GT-HAD [42] (2024)	DFAN [38] (2024)	DPMN (Ours)
HYDICE	$AUC_{(D,F)}\uparrow$		0.9910	0.9970	0.9791	0.9932	0.8035	0.9828	0.9059	0.9977	**0.9985**
	$AUC_{(D,\tau)}\uparrow$		**0.5700**	0.5287	0.5600	0.1859	0.1885	0.3437	0.1748	0.4500	0.1065
	$AUC_{(F,\tau)}\downarrow$		0.0884	0.0396	0.0673	0.0077	0.0904	0.0083	0.0404	0.0838	**0.0009**
	$AUC_{ODP}\uparrow$		2.4726	**2.4861**	2.4718	2.1714	1.9016	2.3182	2.0403	2.3639	2.1041
	$AUC_{SNPR}\uparrow$		6.4478	13.3510	8.3210	24.1429	2.0852	41.4096	4.3267	5.3699	**118.3333**
	Time (s)↓	Train	41.37	18.95	44.10	34.06	37.35	29.55	58.24	31.46	54.81
		Test				0.0132	0.0371	0.0859	0.1941	0.0265	**0.0122**
Hyperion	$AUC_{(D,F)}\uparrow$		0.9774	0.9935	0.9854	0.9942	0.9413	0.9951	0.9930	0.9978	**0.9994**
	$AUC_{(D,\tau)}\uparrow$		0.3123	0.4355	**0.4478**	0.2217	0.2407	0.2004	0.2775	0.4102	0.0950
	$AUC_{(F,\tau)}\downarrow$		0.1270	0.0556	0.0810	0.0097	0.0416	0.0073	0.0166	0.0804	**0.0004**
	$AUC_{ODP}\uparrow$		2.1627	**2.3734**	2.3522	2.2062	2.1404	2.1882	2.2539	2.3276	2.0940
	$AUC_{SNPR}\uparrow$		2.4591	7.8327	5.5284	22.8557	5.7861	27.4521	16.7169	5.1020	**237.5000**
	Time (s)↓	Train	114.12	43.90	86.33	17.76	87.54	89.43	110.50	25.70	102.22
		Test				**0.0125**	0.0454	0.0636	0.3683	0.1412	0.0141
ABU-beach-1	$AUC_{(D,F)}\uparrow$		0.9546	0.9743	0.9766	0.9509	0.9510	0.9205	0.9906	0.9822	**0.9917**
	$AUC_{(D,\tau)}\uparrow$		0.2671	0.3323	0.1959	0.1242	0.1288	0.2529	0.2701	**0.5707**	0.0843
	$AUC_{(F,\tau)}\downarrow$		0.0467	0.0185	0.0088	0.0173	0.0183	0.0281	0.0069	0.0348	**0.0001**
	$AUC_{ODP}\uparrow$		2.1750	2.2881	2.1637	2.0578	2.0615	2.1453	2.2538	**2.5181**	2.0759
	$AUC_{SNPR}\uparrow$		5.7195	17.9622	22.2614	7.1792	7.0385	9.0000	39.1449	16.3994	**843.0000**
	Time (s)↓	Train	72.90	67.70	128.25	27.72	41.32	94.26	78.69	46.41	60.21
		Test				0.0147	0.0416	0.1153	0.5246	0.1430	**0.0119**
ABU-beach-3	$AUC_{(D,F)}\uparrow$		0.9287	0.9947	0.9930	0.9983	0.9276	0.9889	0.9993	0.9989	**0.9995**
	$AUC_{(D,\tau)}\uparrow$		0.5898	0.5663	0.5508	0.3744	0.4743	0.2992	**0.6233**	0.5843	0.1530
	$AUC_{(F,\tau)}\downarrow$		0.2227	0.0498	0.0767	0.0045	0.1141	0.0121	0.0319	0.0701	**0.0004**
	$AUC_{ODP}\uparrow$		2.2958	2.5112	2.4671	2.3682	2.2878	2.2760	2.5907	**2.5131**	2.1521
	$AUC_{SNPR}\uparrow$		2.6484	11.3715	7.1812	83.2000	4.1569	24.7273	19.5391	8.3352	**382.5000**
	Time (s)↓	Train	75.78	36.41	49.75	20.27	23.83	25.48	35.66	15.56	65.11
		Test				**0.0129**	0.0263	0.0865	0.3566	0.0174	0.0146
Pavia	$AUC_{(D,F)}\uparrow$		0.9944	0.9958	0.9943	0.9891	0.9927	0.9944	**0.9992**	0.9955	0.9991
	$AUC_{(D,\tau)}\uparrow$		0.2843	0.2766	0.3184	0.0905	0.2132	0.2004	0.2439	**0.5888**	0.1092
	$AUC_{(F,\tau)}\downarrow$		0.0233	0.0165	0.0240	0.0023	0.0045	0.0059	0.0019	0.0771	**0.0006**
	$AUC_{ODP}\uparrow$		2.2554	2.2559	2.2887	2.0773	2.2014	2.1889	2.2412	**2.5072**	2.1077
	$AUC_{SNPR}\uparrow$		12.2017	16.7636	13.2667	39.3478	47.3778	33.9661	128.3684	7.6368	**182.0000**
	Time (s)↓	Train	40.03	17.92	24.79	5.98	13.57	28.85	38.55	14.99	11.63
		Test				0.0134	**0.0130**	0.1430	0.2570	0.1190	0.0135
Salinas	$AUC_{(D,F)}\uparrow$		0.9962	0.9993	0.9957	0.9951	0..9991	0.9955	0.9988	0.9954	**0.9998**
	$AUC_{(D,\tau)}\uparrow$		**0.5850**	0.5745	0.5302	0.1959	0.4662	0.4013	0.4547	0.4261	0.2258
	$AUC_{(F,\tau)}\downarrow$		0.0379	0.0238	0.0377	0.0026	0.0075	0.0105	0.0060	0.0634	**0.0008**
	$AUC_{ODP}\uparrow$		2.5433	**2.5500**	2.4882	2.1884	2.4578	2.3863	2.4475	2.3581	2.2248
	$AUC_{SNPR}\uparrow$		15.4354	24.1387	14.0637	75.3462	62.1600	38.2190	75.7833	6.7208	**282.2500**
	Time (s)↓	Train	92.18	77.22	214.91	39.82	158.84	93.90	115.94	21.98	21.49
		Test				0.0124	0.0958	0.0369	0.7729	0.1874	**0.0098**
AVIRIS-1	$AUC_{(D,F)}\uparrow$		0.9961	0.9910	0.9928	0.9915	0.9849	**0.9967**	0.9957	0.9902	0.9963
	$AUC_{(D,\tau)}\uparrow$		0.5432	0.5195	0.4225	0.2489	0.3318	0.3810	**0.6063**	0.4356	0.1430
	$AUC_{(F,\tau)}\downarrow$		0.0962	0.0541	0.0550	0.0203	0.0427	0.0159	0.0569	0.0996	**0.0077**
	$AUC_{ODP}\uparrow$		2.4431	2.4564	2.3603	2.2201	2.2740	2.3618	**2.5451**	2.3262	2.1316
	$AUC_{SNPR}\uparrow$		5.6466	9.6026	7.6819	12.2611	7.7705	**23.9623**	10.6555	4.3735	18.5714
	Time (s)↓	Train	61.97	20.20	46.91	6.88	47.04	38.62	36.74	15.80	32.61
		Test				0.0558	0.0501	0.0127	0.2449	0.6999	**0.0119**
AVIRIS-2	$AUC_{(D,F)}\uparrow$		0.9909	0.9941	0.9918	0.9609	0.9957	0.9952	0.9946	0.9953	**0.9971**
	$AUC_{(D,\tau)}\uparrow$		0.5250	0.3908	0.4762	0.1292	0.2864	0.2842	0.4331	**0.5489**	0.3042
	$AUC_{(F,\tau)}\downarrow$		0.1022	0.0270	0.0494	**0.0067**	0.0254	0.0100	0.0207	0.1076	0.0074
	$AUC_{ODP}\uparrow$		2.4137	2.3579	2.4186	2.0834	2.2567	2.2694	2.4070	**2.4366**	2.2939
	$AUC_{SNPR}\uparrow$		5.1370	14.4741	9.6397	19.2836	11.2756	28.4200	20.9227	5.1013	**41.1081**
	Time (s)↓	Train	62.72	20.12	46.72	33.60	42.77	39.15	35.68	15.34	25.10
		Test				0.0105	0.0421	0.0743	0.2378	0.1539	**0.0104**

Moreover, to quantitatively demonstrate the effectiveness and necessity of different components of the proposed method, ablation studies are conducted, and $AUC_{(D,F)}$ is used to reflect the comprehensive performance of different detectors. Since the generative module serves as the baseline of the entire network, ablation experiments are conducted solely on

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TABLE III

DETECTION ACCURACY OF THE PROPOSED METHOD WITH DIAGONAL DIRECTION ON THE EIGHT DATASETS

Datasets	AUC Values		HYDICE	Hyperion	ABU-beach-1	ABU-beach-3	Pavia	Salinas	AVIRIS-1	AVIRIS-2
W/ diagonal	$AUC_{(D,F)}\uparrow$		**0.9986**	0.9993	0.9915	**0.9997**	**0.9991**	**0.9998**	**0.9972**	0.9950
	Time (s) ↓	Train	764.49	1001.57	2241.20	881.06	361.04	1926.20	531.92	767.76
		Test	0.5343	1.5678	1.6147	0.6623	0.7186	2.5089	0.2396	0.7961
W/O diagonal	$AUC_{(D,F)}\uparrow$		0.9985	**0.9994**	**0.9917**	0.9995	**0.9991**	**0.9998**	0.9963	**0.9971**
	Time (s) ↓	Train	**54.81**	**102.22**	**60.21**	**65.11**	**11.63**	**21.49**	**32.61**	**25.10**
		Test	**0.0122**	**0.0141**	**0.0019**	**0.0146**	**0.0135**	**0.0098**	**0.0119**	**0.0104**

TABLE IV

$AUC_{(D,F)}$ OF THE PROPOSED METHOD ON DIFFERENT ABLATION CASES

Dictionary-Learnable Block	Spatial Smoothing Filter	Merging TV into LRR	HYDICE	Hyperion	ABU-beach-1	ABU-beach-3	Pavia	Salinas	AVIRIS-1	AVIRIS-2	Average
×	√	√	0.9966	0.9977	0.9669	0.9978	0.9957	0.9979	0.9896	0.9914	0.9917
√	×	√	0.9921	0.9978	0.9785	0.9983	0.9959	0.9942	0.9929	0.9935	0.9929
√	√	×	0.9960	0.9969	0.9895	0.9975	0.9987	0.9993	0.9944	0.9937	0.9956
√	√	√	**0.9985**	**0.9994**	**0.9917**	**0.9995**	**0.9991**	**0.9998**	**0.9963**	**0.9971**	**0.9977**

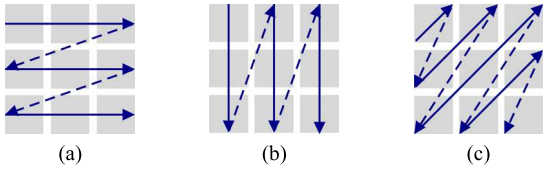


Fig. 9. Different sequence generation directions. (a) Horizontal direction. (b) Vertical direction. (c) Diagonal direction.

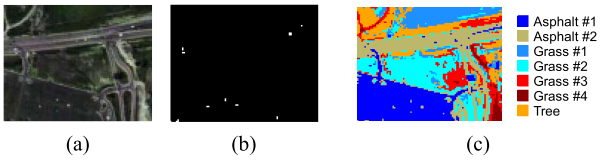


Fig. 10. Clustering results on the HYDICE dataset. (a) Pseudo-color image. (b) Ground truth. (c) Clustering result.

the dictionary-learnable block, spatial smoothing filter, and loss function. The initial background dictionary is obtained through the DPC algorithm, taking the HYDICE dataset as an example. Fig. 10 illustrates the clustering result of DPC on the HYDICE dataset, identifying a total of seven background classes. According to [61], among these seven categories, two are asphalt, four are grass, and the remaining one is tree, while each anomalous target is assigned to the background class closest to it. To address the inaccuracy of the initial background dictionary and enhance the network flexibility, we incorporated the dictionary-learnable block. From Table IV, it is demonstrated that the dictionary-learnable block can enhance detection accuracy across all datasets. Moreover, it can also be observed from Table IV that adding a spatial smoothing filter before output enhances the detection accuracy since it aids the network in better capturing details and edge information. Furthermore, incorporating the prior physical knowledge of HSIs into the loss function can enhance detection accuracy because TV and LRR can effectively mitigate the impact of noise and complex textures within the scenes. Therefore, the network design of the proposed method is justified.

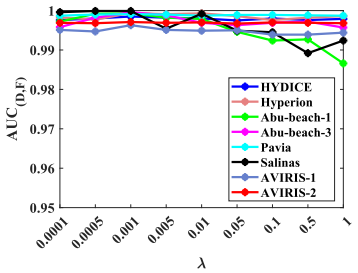


Fig. 11. Detection accuracy of the proposed method on the eight datasets varies with the value of λ.

2) Hyperparameter Analysis: By incorporating a regularization term that merges TV into LRR into the loss function, the proposed method leverages both the spatial smoothness and low-rank properties of HSIs, with introducing only one additional hyperparameter λ. To better illustrate the impact of parameter variations on detection results, the optimal value of λ is selected from the provided range [1, 0.0001,0.0005,0.001,0.005,0.01,0.05,0.1,0.5]. As shown in Fig. 11, when λ is less than 0.05, the detection accuracy of DPMN begins to exhibit slight fluctuations and declines across all datasets, indicating that the proposed method demonstrates insensitivity to λ values within the range of [0.0001, 0.05]. Therefore, λ is set to 0.001 in the experiments and applied to all datasets.

3) Analysis of the Impact of the Number of Atoms in the Background Dictionary: The number of atoms in the background dictionary determines the number of input channels to the network, which is decided by the number of clusters in the DPC algorithm. In the DPC algorithm, the number of clusters is determined by both the parameter α and β. The setting of α impacts the choice of truncation distance d_c, thereby affecting the calculation of local density for each pixel. While, β controls the selection of cluster center pixels. Fig. 12 shows the variation in the number of clusters and $AUC_{(D,F)}$ of the proposed method with different α and β values on the HYDICE dataset. The experimental results indicate that the detection

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TABLE V

$AUC_{(D,F)}$ VALUES (MEAN ± STANDARD DEVIATION) WERE RUN 20 TIMES ON THE HYDICE DATASET WITH DIFFERENT LEVELS OF ZERO-MEAN GAUSSIAN NOISE FOR DIFFERENT METHODS

Methods	PCA-TLRSR [30] (2023)	LARTVAD [31] (2023)	TRDFTVAD [32] (2023)	Auto-AD [34] (2022)	RGAE [35] (2022)	GAED [36] (2022)	GT-HAD [42] (2024)	DFAN [38] (2024)	DPMN (Ours)
30 dB	0.9434±0.0285	0.9412±0.0270	0.9440±0.0108	0.9918±0.0022	0.7909±0.0050	0.9467±0.0216	0.8974±0.0056	0.9926±0.0024	**0.9943±0.0013**
25 dB	0.9127±0.0315	0.9274±0.0391	0.9209±0.0257	0.9766±0.0076	0.7706±0.0142	0.9136±0.0295	0.8967±0.0060	0.9699±0.0062	**0.9779±0.0047**
20 dB	0.8580±0.0369	0.8007±0.0639	0.8507±0.0498	0.9060±0.0333	0.7414±0.0277	0.8486±0.0479	0.8769±0.0082	0.9244±0.0096	**0.9328±0.0067**

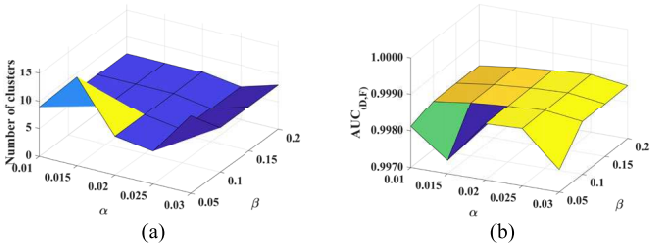


Fig. 12. Number of clusters and $AUC_{(D,F)}$ of the proposed method on the HYDICE datasets vary with the value of α and β. (a) Variation in the number of clusters. (b) Variation in the value of $AUC_{(D,F)}$.

accuracy of the proposed method remains highly stable over a wide range of α and β values. In all experiments conducted in this article, the α and β values are set to 0.02 and 0.1, respectively, as recommended in [58] and [59]. Therefore, α and β are not treated as a hyperparameter to be tuned in the proposed method.

4) Noise Robustness Evaluation: Noise robustness experiments are conducted on the HYDICE dataset, which features a complex textured scene. First, various levels of Gaussian noise are added to the HYDICE dataset to generate a noisy experimental dataset, where the noise level is measured by the SNR, which can be calculated as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{\mathbb{E}\left[\boldsymbol{y}^T\boldsymbol{y}\right]}{\mathbb{E}\left[\boldsymbol{e}^T\boldsymbol{e}\right]} \tag{21}$$

where $\boldsymbol{y}$ and $\boldsymbol{e}$ denote the original data and the additive noise, respectively. $\mathbb{E}[\cdot]$ stands for calculating the expectation. Second, to minimize randomness, 20 repeated experiments are conducted at each noise level, with independent Gaussian noise randomly added each time. Table V displays the mean and standard deviation of the $AUC_{(D,F)}$ for the detection results of different detectors under each noise level, from which it can be seen that the proposed DPMN achieves the best detection accuracy and superior noise robustness. This could be attributed to the embedding of physical knowledge into the loss function, along with the introduction of the learnable dictionary, which effectively mitigates the impact of noise, thereby enhancing the noise robustness and performance stability of the proposed method.

E. Analysis of Complexity and Convergence

1) Complexity Analysis: We record the number of model parameters and floating point operations (FLOPs) metrics of five deep learning-based methods in detail on the HYDICE datasets, which are used to assess model size and computational complexity, respectively, as shown in Table VI. As shown in Table VI, our method significantly reduces the number of parameters compared to Auto-AD. Although the proposed method has higher FLOPs, its efficiency is improved thanks to the parallel scanning algorithm of Mamba. As evidenced by Table II, our method achieves relatively short test times.

TABLE VI

FIVE DEEP LEARNING-BASED METHODS' PARAMETERS (MILLIONS) AND FLOPS (BILLIONS) ON THE HYDICE DATASET

Methods	Auto-AD	RGAE	GAED	GT-HAD	DFAN	DPMN
Params (M)	3.21	0.03	0.04	0.22	0.07	1.55
FLOPs (B)	4.65	0.31	0.11	1.03	0.52	6.30

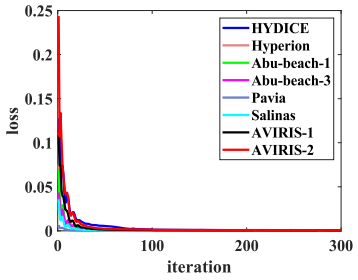


Fig. 13. Training loss curves for different datasets.

2) Convergence Analysis: To demonstrate the convergence of the proposed method, Fig. 13 presents the training loss curves for the eight real datasets. It can be seen that the training loss initially drops sharply and then gradually approaches 0 as the number of training iterations increases, thereby confirming the convergence of the proposed DPMN.

F. Insightful Analysis

The proposed method mainly consists of the AGM module and the BRM module. It generates background abundances through the AGM module and combines them with the learnable background dictionary extracted by the DPC algorithm in the BRM module to reconstruct the background. Since the background is divided into different subspaces for reconstruction, and both local and global information are fully utilized, the background can be effectively reconstructed while the interference to the background reconstruction caused by anomalies is reduced. Thus, the proposed DPMN achieves a significant performance improvement in handling HAD tasks.

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Based on the above experimental results, we provide the following discussions and insightful analysis.

1) *Effectiveness of Detection:* The proposed method not only leverages the convolutional network to extract local information but also introduces the bidirectional Mamba block to extract global information. Moreover, by incorporating a learnable background dictionary, the proposed method not only projects the background into distinct subspaces for reconstruction, thereby achieving more accurate background reconstruction but also reduces the risk of anomalous targets being reconstructed, as the anomalies cannot be represented by the background dictionary. Furthermore, incorporating a regularization term that merges TV into LRR in the loss function guides the network to accurately reconstruct the background. The experimental results on the eight real datasets demonstrate that our method achieves superior performance even in handling anomalous targets of varying sizes.
2) *Efficiency of Detection:* The experimental results have verified that although the methods based on deep learning require longer training times, their testing times are significantly shorter than those of the physics-based methods solved by iterative optimization. It is worth noting that although the proposed method involves relatively long training times, the test (inferring) time is comparatively short, indicating that the proposed DPMN should be applicable for real-time anomaly detection.
3) *Robustness to Noise:* We validated the noise robustness of the proposed DPMN method by adding Gaussian noise of varying degrees to the HYDICE dataset. This is primarily due to incorporating prior knowledge of HSIs into the network, which enhances noise robustness. Specifically, by smoothing the data with TV to significantly suppress noise and using LRR to better extract the main features and information from the data, the difference between noise and the intrinsic structure of the data is notably enhanced. Moreover, with the help of a continuously updated learnable background dictionary, DPMN can learn more precise background features and effectively eliminate the adverse effects of noise and interference in the background, achieving stable and accurate background reconstruction.
4) *Convenience of Parameter Setting:* The proposed DPMN features a single hyperparameter λ in its loss function. Parameter experiments across the eight datasets demonstrate that the detection performance of DPMN is insensitive to λ values within a relatively wide range. Moreover, the superior detection results of DPMN are achieved when λ is fixed at 0.001 across all experiments.
5) *Limitations:* In real-world scenarios, factors such as the complex distribution of ground objects and terrain lead to interactions between light and multiple objects before reaching the sensor, resulting in nonlinear spectral mixing phenomena in HSIs. Since the abundances and dictionary generated by the BRM are combined through a linear layer to reconstruct the background, the proposed method at present may not accurately recover the background of HSIs that exhibit nonlinear spectral mixing phenomena in reality. In future work, we plan to embed concrete nonlinear spectral mixing models within the BRM, while keeping the AGM and loss function unchanged, which is expected to enhance the ability of the proposed method to handle nonlinear HSIs.

V. Conclusion

In this article, we have proposed a novel approach named DPMN for HAD. In contrast to the current deep learning-based HAD methods, the proposed approach integrates CNN and bidirectional Mamba into AGM to more effectively capture both local and global information, thereby enabling precise extraction of background features. In addition, BRM divides the background into different subspaces by incorporating a learnable background dictionary that can be updated through network learning, facilitating more accurate background reconstruction. Furthermore, our method introduces a regularization term that merges TV into LRR and incorporates it into the loss function to guide background reconstruction, leveraging the prior physical knowledge of HSIs to enhance the model's generalization capability and improve detection accuracy. The experimental results on the eight real-world datasets demonstrate that the proposed method not only excels in high detection accuracy with relatively short inference time across varying sizes of anomalous targets but also offers the advantages of convenient parameter setting and strong robustness against noise.

In future work, we will consider embedding specific nonlinear spectral mixing models into BRM to adapt the proposed method for handling nonlinear HAD in real-world scenarios.

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